

## Assignment 4

Deadline: 11/17/2025, h: 11:59pm

- This assignment contains 3 exercises. Submit your solution via Gradescope. See the syllabus for the late submission policy.
- You are allowed to discuss the assignment with others but the write-up must be individual work. Please mention in your write-up all the people you have discussed the solution with.

**Problem 1.** Consider the following project management problem, similar to the one seen in class. Suppose that we are given a sequence of activities  $1, \dots, n$ , where each activity  $i$  has an *execution time*  $c_i$  and a list of *prerequisite* activities that need to be completed before activity  $i$  is started. The goal is to find the *project completion time*, that is, the time required to complete all activities, assuming that we can perform in parallel any set of activities that are not prerequisite of one another.

An example of this problem is given below.

| activity     | 1    | 2    | 3 | 4 | 5   | 6   | 7   | 8   | 9   | 10  |
|--------------|------|------|---|---|-----|-----|-----|-----|-----|-----|
| $c_i$        | 2    | 5    | 4 | 1 | 3   | 7   | 9   | 6   | 3   | 2   |
| prerequisite | none | none | 1 | 2 | 3,4 | 3,5 | 4,6 | 2,7 | 3,6 | 8,9 |

- Formulate the problem of finding the project completion time as a longest path problem in an appropriate digraph;
- Formulate the previous problem as a shortest path problem in an appropriate digraph;
- Formulate the previous problem as an instance of the min-cost flow problem, and compute its optimal optimal solution by solving a linear program (do not submit the code).
- We say that an activity is in a *critical path* if any delay in its completion will cause the completion time of the project to be delayed. Give an algorithm to compute which activities are in the critical path (no need to run it; just state the algorithm).

**Problem 2:** Your company sells phones and owns one warehouse and one shop. In days 1,2,3, the shop needs to satisfy a given demand  $d_1, d_2, d_3 > 0$  of phones, respectively. For  $j \in \{1, 2, 3\}$ , phones can be shipped from the warehouse to the shop in day  $j$  at a given per-unit cost  $c_j > 0$ . For  $j \in \{1, 2\}$ , phones already at the shop can be stored between day  $j$  and  $j + 1$  at a given per-unit cost  $q_j > 0$ . Assume that the warehouse has enough capacity as to satisfy the demands of the shop.

- Formulate the problem of shipping the phones from the warehouse to the shop in the different days as to satisfy the demand at minimum cost as an integer program.
- Give an integer programming formulation for the following modification of the problem from part 1.: in day 2, the storing cost  $q_2(x)$  is piecewise linear: in particular, the per-unit storing cost is given as follows

$$q_2 = \begin{cases} 10 & \text{if \# stored units} \leq 10 \\ 8 & \text{for every stored unit between 11 and 14} \\ 5 & \text{for every additional stored unit} \end{cases}$$

- Give an integer programming formulation for the following modification of the problem from part 1.: We can decide to rent additional storage space in day 1. If we do so, we pay a fixed cost of  $k$ , but the per-unit storing cost  $q_1$  is divided by 3. Else, we pay for storing costs as in part 1.

**Problem 3:** Consider the following modification of the problem seen in the recitation on 11/07. Your shop is open from 8am to 6pm. Based on past customer data, you know that to serve your customers you will need a different number of clerks and supervisors during the day:

| hour          | 8-9 | 9-10 | 10-11 | 11-12 | 12-1 | 1-2 | 2-3 | 3-4 | 4-5 | 5-6 | 6-7 |
|---------------|-----|------|-------|-------|------|-----|-----|-----|-----|-----|-----|
| # clerks      | 3   | 4    | 2     | 5     | 7    | 8   | 10  | 3   | 4   | 5   | 5   |
| # supervisors | 1   | 1    | 1     | 1     | 2    | 2   | 3   | 1   | 1   | 1   | 1   |

Clerks must be hired for a consecutive shift of 3 hours, while supervisors for a consecutive shift of 2 hours.

- Formulate as an integer program the problem of finding the minimum total number of clerks and supervisors to hire so that you will be able to serve your customers.
- Suppose now a clerk is paid 19 dollars per hour, while a supervisor is paid 27 dollars per hour. Now your goal is to minimize the total salary you pay, subject to the constraints from part a). Does the optimal solution change? Explain.
- Now suppose that every worker who would work at least one hour before 11am gets a 10% “early bird” salary bonus (over their full salary), and every worker who finishes working at 5pm or later gets a 20% “late shift” salary bonus (on their full salary). Formulate the problem of minimizing the total salary paid, subject to the constraints from part a) and the base salary from part b).